

MYSQL VIEWS

Practical Examples and Student Exercises using MySQL Workbench

1.What is a View?

A **View** is a **virtual table** created from one or more existing tables.

- It **does not store data** itself.
- It stores only the **SQL query**.
- Whenever you query the view, MySQL retrieves the latest data from the underlying tables.

2. Setup: Employees and Departments Tables

Run this setup first. If these tables already exist with data, you may skip the setup.

```
CREATE DATABASE IF NOT EXISTS employeedb1;  
USE employeedb1;
```

```
CREATE TABLE IF NOT EXISTS departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE IF NOT EXISTS employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    salary DECIMAL(10,2)  
);
```

```
INSERT IGNORE INTO departments VALUES  
(1,'CSE'),  
(2,'ECE'),  
(3,'EEE');
```

```
INSERT IGNORE INTO employees VALUES  
(101,'Rahul',1,50000),  
(102,'Priya',2,65000),  
(103,'Anil',1,55000),  
(104,'Sneha',3,70000),  
(105,'Kiran',2,48000);
```

3. Verify the Tables

SHOW TABLES;

SELECT * FROM departments;

SELECT * FROM employees;

View 1 – Display All Employees

Question: Create a View named EmployeeView to display all columns from employees.

```
7 • CREATE VIEW EmployeeView AS
8   SELECT * FROM employees;
9 • SELECT * FROM EmployeeView;
10
11
```

Result Grid						
Filter Rows: Export: Wrap Cell Conte						
	EmpID	EmpName	Salary	DeptID	ManagerID	Email
▶	101	Rahul	55000.00	1	NULL	rahul@gmail.com
	102	Anjali	65000.00	2	101	anjali@gmail.com
	103	Rohan	49500.00	1	101	rohan@gmail.com
	105	Kiran	77000.00	1	101	kiran@gmail.com
	106	Sneha	62000.00	4	105	sneha@gmail.com
	107	Arjun	32000.00	NULL	101	arjun@gmail.com
	108	Vijay	58000.00	2	102	vijay@gmail.com

View 2 – Selected Columns

Question: Create a View showing employee ID, employee name and salary only.

```
10 • CREATE VIEW EmployeeBasicView AS
11   SELECT EmpID, EmpName, Salary
12   FROM employees;
13 • SELECT * FROM EmployeeBasicView;
```

Result Grid			
Filter Rows: Exp			
	EmpID	EmpName	Salary
▶	101	Rahul	55000.00
	102	Anjali	65000.00
	103	Rohan	49500.00
	105	Kiran	77000.00
	106	Sneha	62000.00
	107	Arjun	32000.00
	108	Vijay	58000.00

3 – High Salary Employees

Question: Create a View that displays employees earning more than 55000.

```
14 • CREATE VIEW HighSalaryEmployees AS
15 SELECT EmpID, EmpName, Salary
16 FROM employees
17 WHERE salary > 55000;
18 • SELECT * FROM HighSalaryEmployees;
```

Result Grid				
	EmpID	EmpName	Salary	
▶	102	Anjali	65000.00	
	105	Kiran	77000.00	
	106	Sneha	62000.00	
	108	Vijay	58000.00	

View 4 – Employee and Department using JOIN

Question: Create a View that displays employee name, department name and salary.

```
20 SELECT
21 e.EmpID,
22 e.EmpName,
23 d.DeptName,
24 e.Salary
25 FROM employees e
26 JOIN departments d
27 ON e.DeptID = d.DeptID;
28
29 • SELECT * FROM EmployeeDepartmentView;
```

Result Grid				
	EmpID	EmpName	DeptName	Salary
▶	101	Rahul	CSE	55000.00
	103	Rohan	CSE	49500.00
	105	Kiran	CSE	77000.00
	102	Anjali	ECE	65000.00
	108	Vijay	ECE	58000.00
	106	Sneha	MBA	62000.00

View 5 – Employees from a Particular Department

Question: Create a View that displays only CSE employees.

```
3 • CREATE view CSEEmployees AS
4 SELECT
5     e.EmpID,
6     e.EmpName,
7     d.DeptName,
8     e.Salary
9 FROM employees e
10 JOIN departments d
11 ON e.DeptID = d.DeptID
12 WHERE d.DeptName = 'CSE';
13
14 • SELECT * FROM CSEEmployees;
15
```

Result Grid | Filter Rows: | Export:

EmpID	EmpName	DeptName	Salary
101	Rahul	CSE	55000.00
103	Rohan	CSE	49500.00
105	Kiran	CSE	77000.00

View 6 – Department-wise Average Salary

Question: Create a View using GROUP BY and AVG() to display department-wise average salary.

```
14
15 • CREATE VIEW DepartmentSalaryView1 AS
16 SELECT
17     d.DeptName,
18     AVG(e.Salary) AS AverageSalary
19 FROM Employees e
20 JOIN Departments d
21 ON e.DeptID = d.DeptID
22 GROUP BY d.DeptID, d.DeptName;
23
24 • SELECT * FROM DepartmentSalaryView1;
```

Result Grid | Filter Rows: | Export:

DeptName	AverageSalary
CSE	60500.000000
ECE	61500.000000
MBA	62000.000000

View 7 – Departments with Average Salary Above 55000

Question: Create a View that displays departments whose average salary is greater than 55000.

```
27 • CREATE VIEW HighAverageDepartments AS
28 SELECT
29     d.DeptName,
30     AVG(e.Salary) AS AverageSalary
31 FROM employees e
32 JOIN departments d
33 ON e.DeptID = d.DeptID
34 GROUP BY d.DeptID, d.DeptName
35 HAVING AVG(e.Salary) > 55000;
36
37 • SELECT * FROM HighAverageDepartments;
38
```

Result Grid			Filter Rows:	Export:
	DeptName	AverageSalary		
▶	CSE	60500.000000		
	ECE	61500.000000		
	MBA	62000.000000		

View 8 – Salary Range View

Question: Create a View that displays employees with salary between 50000 and 70000.

```
40 • CREATE VIEW SalaryRangeView AS
41 SELECT EmpID, EmpName, Salary
42 FROM employees
43 WHERE Salary BETWEEN 50000 AND 70000;
44
45 • SELECT * FROM SalaryRangeView;
46
```

Result Grid			Filter Rows:	Export:
	EmpID	EmpName	Salary	
▶	101	Rahul	55000.00	
	102	Anjali	65000.00	
	106	Sneha	62000.00	
	108	Vijay	58000.00	

View 9 – View with Calculated Column

Question: Create a View that displays employee name, monthly salary and annual salary.

```
47 • CREATE VIEW AnnualSalaryView AS
48 SELECT
49     EmpID,
50     EmpName,
51     Salary AS MonthlySalary,
52     Salary * 12 AS AnnualSalary
53 FROM employees;
54
55 • SELECT * FROM AnnualSalaryView;
56
```

Result Grid				
Filter Rows: Export: Wr				
EmpID	EmpName	MonthlySalary	AnnualSalary	
101	Rahul	55000.00	660000.00	
102	Anjali	65000.00	780000.00	
103	Rohan	49500.00	594000.00	
105	Kiran	77000.00	924000.00	
106	Sneha	62000.00	744000.00	
107	Arjun	32000.00	384000.00	
108	Vijay	58000.00	696000.00	

View 10 – Update Data Through a Simple View

Question: Create a simple single-table View and update an employee's salary through the View.

```
57 • CREATE VIEW EmployeeSalaryView AS
58 SELECT EmpID, EmpName, Salary
59 FROM employees;
60
61 • UPDATE EmployeeSalaryView
62 SET Salary = 60000
63 WHERE EmpID = 101;
64
65 • SELECT * FROM EmployeeSalaryView;
66 • SELECT * FROM employees WHERE EmpID = 101;
67
```

Result Grid					
Filter Rows: Edit: Export					
EmpID	EmpName	Salary	DeptID	ManagerID	Email
101	Rahul	60000.00	1	NULL	rahul@gmail.com
•	NULL	NULL	NULL	NULL	NULL

View 11 – CREATE OR REPLACE VIEW

Question: Modify an existing View so that it displays employee name and salary only.

```
68 • CREATE OR REPLACE VIEW EmployeeBasicView AS
69 SELECT EmpName, Salary
70 FROM employees;
71
72 • SELECT * FROM EmployeeBasicView;
73
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

	EmpName	Salary
▶	Rahul	60000.00
	Anjali	65000.00
	Rohan	49500.00
	Kiran	77000.00
	Sneha	62000.00
	Arjun	32000.00
	Vijay	58000.00

View 12 – View Definition

Question: Display the SQL statement used to create a View.

```
SHOW CREATE VIEW EmployeeDepartmentView1;
```

View 13 – List All Views

Question: Display all Views available in the current database.

```
76 • SHOW FULL TABLES
77 WHERE Table_type = 'VIEW';
78
```

Result Grid | Filter Rows: | Export:

	Tables_in_company_db	Table_type
▶	annualsalaryview	VIEW
	cseemployees	VIEW
	departmentaveragesalary	VIEW
	departmentaveragesalary1	VIEW
	departmentsalaryview	VIEW
	employeebasicview	VIEW
	employeesalaryview	VIEW
	employeeview	VIEW
	highaveragedepartments	VIEW
	highsalaryemployees	VIEW
	salaryrangeview	VIEW

View 14 – Drop a View

Question: Remove an existing View from the database.

```
DROP VIEW IF EXISTS SalaryRangeView;
```